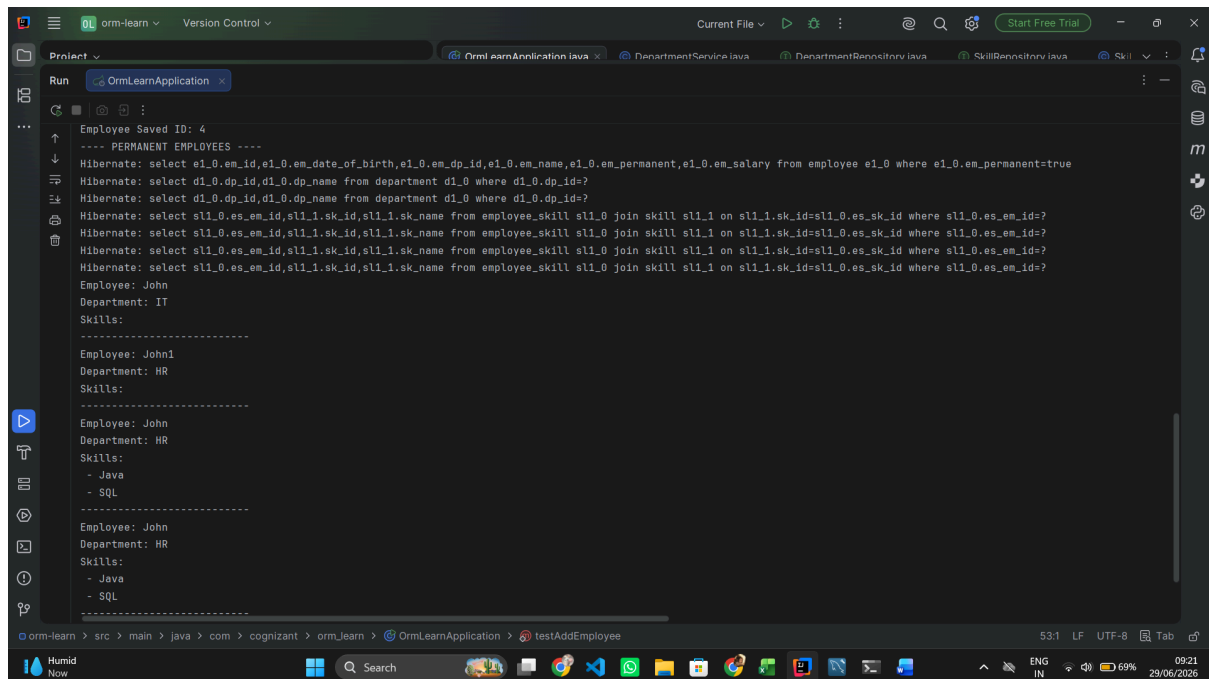


Output:

output:



```
Employee Saved ID: 4
---- PERMANENT EMPLOYEES ----
Hibernate: select e1_0.em_id,e1_0.em_date_of_birth,e1_0.em_dp_id,e1_0.em_name,e1_0.em_permanent,e1_0.em_salary from employee e1_0 where e1_0.em_permanent=true
Hibernate: select d1_0.dp_id,d1_0.dp_name from department d1_0 where d1_0.dp_id=?
Hibernate: select d1_0.dp_id,d1_0.dp_name from department d1_0 where d1_0.dp_id=?
Hibernate: select s1_1.es_em_id,s1_1.sk_id,s1_1.sk_name from employee_skill s1_0 join skill s1_1 on s1_1.sk_id=s1_0.es_sk_id where s1_0.es_em_id=?
Hibernate: select s1_1.es_em_id,s1_1.sk_id,s1_1.sk_name from employee_skill s1_0 join skill s1_1 on s1_1.sk_id=s1_0.es_sk_id where s1_0.es_em_id=?
Hibernate: select s1_1.es_em_id,s1_1.sk_id,s1_1.sk_name from employee_skill s1_0 join skill s1_1 on s1_1.sk_id=s1_0.es_sk_id where s1_0.es_em_id=?
Employee: John
Department: IT
Skills:
-----
Employee: John1
Department: HR
Skills:
-----
Employee: John
Department: HR
Skills:
- Java
- SQL
-----
Employee: John
Department: HR
Skills:
- Java
- SQL
-----
```

src/main/resources/application.properties

```
spring.application.name=orm-learn

spring.datasource.url=jdbc:mysql://localhost:3306/ormlearn
spring.datasource.username=root
spring.datasource.password=997662

spring.jpa.hibernate.ddl-auto=validate
spring.jpa.show-sql=true
```

src/main/java/com/cognizant/orm_learn/model/Employee.java

```
package com.cognizant.orm_learn.model;

import jakarta.persistence.*;

import java.util.Date;
import java.util.List;

@Entity
@Table(name = "employee")
public class Employee {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    @Column(name = "em_id")
```

```

private int id;

@Column(name = "em_name")
private String name;

@Column(name = "em_salary")
private double salary;

@Column(name = "em_permanent")
private boolean permanent;

@Temporal(TemporalType.DATE)
@Column(name = "em_date_of_birth")
private Date dateOfBirth;

@ManyToOne
@JoinColumn(name = "em_dp_id")
private Department department;

@ManyToMany(fetch = FetchType.EAGER)
@JoinTable(
    name = "employee_skill",
    joinColumns = @JoinColumn(name = "es_em_id"),
    inverseJoinColumns = @JoinColumn(name = "es_sk_id")
)
private List<Skill> skillList;

public int getId() {
    return id;
}

public void setId(int id) {
    this.id = id;
}

public String getName() {
    return name;
}

public void setName(String name) {
    this.name = name;
}

public double getSalary() {
    return salary;
}

public void setSalary(double salary) {
    this.salary = salary;
}

public boolean isPermanent() {
    return permanent;
}

```

```

    }

    public void setPermanent(boolean permanent) {
        this.permanent = permanent;
    }

    public Date getDateOfBirth() {
        return dateOfBirth;
    }

    public void setDateOfBirth(Date dateOfBirth) {
        this.dateOfBirth = dateOfBirth;
    }

    public Department getDepartment() {
        return department;
    }

    public void setDepartment(Department department) {
        this.department = department;
    }

    public List<Skill> getSkillList() {
        return skillList;
    }

    public void setSkillList(List<Skill> skillList) {
        this.skillList = skillList;
    }
}

```

src/main/java/com/cognizant/orm_learn/model/Department.java

```

package com.cognizant.orm_learn.model;

import jakarta.persistence.*;

@Entity
@Table(name = "department")
public class Department {

    @Id
    @Column(name = "dp_id")
    private int id;

    @Column(name = "dp_name")
    private String name;

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }
}

```

```

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}

```

src/main/java/com/cognizant/orm_learn/model/Skill.java

```

package com.cognizant.orm_learn.model;

import jakarta.persistence.*;

@Entity
@Table(name = "skill")
public class Skill {

    @Id
    @Column(name = "sk_id")
    private int id;

    @Column(name = "sk_name")
    private String name;

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }
}

```

src/main/java/com/cognizant/orm_learn/repository/EmployeeRepository.java

```

package com.cognizant.orm_learn.repository;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.data.jpa.repository.Query;

import com.cognizant.orm_learn.model.Employee;

```

```

public interface EmployeeRepository extends JpaRepository<Employee, Integer>
{
    @Query("SELECT e FROM Employee e " +
           "LEFT JOIN FETCH e.department " +
           "LEFT JOIN FETCH e.skillList " +
           "WHERE e.permanent = true")
    List<Employee> findByPermanentTrue();
}

```

src/main/java/com/cognizant/orm_learn/repository/**DepartmentRepository.java**

```

package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.cognizant.orm_learn.model.Department;

public interface DepartmentRepository extends JpaRepository<Department,
Integer> {
}

```

src/main/java/com/cognizant/orm_learn/repository/**SkillRepository.java**

```

package com.cognizant.orm_learn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import com.cognizant.orm_learn.model.Skill;

public interface SkillRepository extends JpaRepository<Skill, Integer> {
}

```

src/main/java/com/cognizant/orm_learn/service/**EmployeeService.java**

```

package com.cognizant.orm_learn.service;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import jakarta.transaction.Transactional;

import com.cognizant.orm_learn.model.Employee;
import com.cognizant.orm_learn.repository.EmployeeRepository;

@Service
public class EmployeeService {

    @Autowired
    private EmployeeRepository employeeRepository;

    @Transactional
    public Employee get(int id) {
        return employeeRepository.findById(id).get();
    }
}

```

```

    @Transactional
    public void save(Employee employee) {
        employeeRepository.save(employee);
    }

    @Transactional
    public java.util.List<Employee> getAllPermanentEmployees() {
        return employeeRepository.findByPermanentTrue();
    }
}

```

src/main/java/com/cognizant/orm_learn/service/DepartmentService.java

```

package com.cognizant.orm_learn.service;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import jakarta.transaction.Transactional;

import com.cognizant.orm_learn.model.Department;
import com.cognizant.orm_learn.repository.DepartmentRepository;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

@Service
public class DepartmentService {

    private static final Logger LOGGER =
LoggerFactory.getLogger(DepartmentService.class);

    @Autowired
    private DepartmentRepository departmentRepository;

    @Transactional
    public Department get(int id) {
        LOGGER.info("Start");
        return departmentRepository.findById(id).get();
    }

    @Transactional
    public void save(Department department) {
        LOGGER.info("Start");
        departmentRepository.save(department);
        LOGGER.info("End");
    }
}

```

src/main/java/com/cognizant/orm_learn/OrmLearnApplication.java

```

package com.cognizant.orm_learn;

import java.text.SimpleDateFormat;

```

```

import java.util.Arrays;
import java.util.Date;
import java.util.List;

import com.cognizant.orm_learn.service.DepartmentService;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

import com.cognizant.orm_learn.model.Employee;
import com.cognizant.orm_learn.model.Department;
import com.cognizant.orm_learn.model.Skill;
import com.cognizant.orm_learn.service.EmployeeService;
import com.cognizant.orm_learn.repository.SkillRepository;

@SpringBootApplication
public class OrmLearnApplication {

    private static EmployeeService employeeService;
    private static DepartmentService departmentService;
    private static SkillRepository skillRepository;

    public static void main(String[] args) {
        ApplicationContext context =
SpringApplication.run(OrmLearnApplication.class, args);

        employeeService = context.getBean(EmployeeService.class);
        departmentService = context.getBean(DepartmentService.class);
        skillRepository = context.getBean(SkillRepository.class);

        testAddEmployee();
        testGetAllPermanentEmployees();
    }

    private static void testAddEmployee() {
        try {
            Employee emp = new Employee();

            emp.setName("John");
            emp.setSalary(50000);
            emp.setPermanent(true);

            Date date = new SimpleDateFormat("yyyy-MM-dd").parse("1995-05-20");
            emp.setDateOfBirth(date);

            Department dept = departmentService.get(1);
            emp.setDepartment(dept);

            List<Skill> skills = skillRepository.findAllById(Arrays.asList(1,
2));

            emp.setSkillList(skills);

            employeeService.save(emp);

```

```
        System.out.println("Employee Saved ID: " + emp.getId());

    } catch (Exception e) {
        e.printStackTrace();
    }
}

private static void testGetAllPermanentEmployees() {

    System.out.println("---- PERMANENT EMPLOYEES ----");

    List<Employee> employees = employeeService.getAllPermanentEmployees();

    for (Employee e : employees) {
        System.out.println("Employee: " + e.getName());
        System.out.println("Department: " + e.getDepartment().getName());

        System.out.println("Skills:");
        e.getSkillList().forEach(s ->
            System.out.println(" - " + s.getName())
        );

        System.out.println("-----");
    }
}
```